

Multi-file project

Real designs are rarely one file

Top, TB, and linkage

- Name the DUT file and the testbench file explicitly
- Know which module is top for the run
- If the IDE links multiple files into one session

Browser lab

Digital Design and Verification Platform

HomeTools

Home / Tools / Multi-file project

INTERACTIVE TOOL

Multi-file project

Document-aid literacy: **top** file, **+define+** macros, **+incdir+** include paths, and **profiles** in the browser HDL simulator (not a real compile). Starter: `top tb/tb_top.v`
`WIDTH=8` · `incdir include` — LINKED.

Starter example: `top tb/tb_top.v` · `+define+WIDTH=8` · `+incdir+include` — LINKED.

Load starter example

Challenges 0 / 22 cleared

Quiz: **top**: The simulation top file is...

☐ the entry point the simulator elaborates first (usually `tb/*.v`)

☐ always the smallest `.v` file

☐ only used for synthesis

☐ the GTKWave config

Show hint

Check

Next

Idle

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

20

21

22

Core ideas

Top

Simulation entry file (TB).

+define+

Preprocessor macros.

+incdir+

include search paths for headers.

Profile

Saved project combo.

Lab

SCENARIO

PROFILE

Lab starter

learn_hdl_simulator — hdl sim multi file

Public simulator practice

- In the public IDE, open a tiny two-file sketch: DUT plus testbench
- Confirm both appear under Files, set top correctly, and elaborate or run once
- Intentionally remove the testbench from the set

Pitfalls to watch

- Do not assume the open editor tab is the top module
- Do not forget packages or included files that the DUT needs
- Do not mix two unrelated testbenches in one run and expect clean waves
- Always reconcile Files with Hierarchy after a change

Your turn

- Complete the checklist for at least one track, preferably both
- Build or restore a coherent DUT-plus-testbench file set and name the top
- When you are ready, take the short quiz, then continue to golden compare